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COMMUNITY ISSUE REPORTING SYSTEM

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I. INTRODUCTION

- **Project Overview**

The Community Issue Reporting System is a digital platform designed to help people report barangay problems in their community area easily. It focuses on improving how communities handle common issues and making the reporting process of the problem more organized and accessible for the barangay to handle and solve the problem as fast as possible.

Users can register or log in and submit reports about community problems like illegal garbage disposal, road damage, flooding, etc. Each report includes details of the problem such as description, category, location, and an optional photo for the prof. The system helps barangay staff manage these reports by assigning priority levels and tracking their status. It also includes features like a history board, automatic sorting, notifications, and a search function. Admins can view all reports through a dashboard and see basic analytics. This project also supports SDG 11 "Sustainable Cities and Communities" which aims to make communities safer and more organized.

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The project helps improve communication between residents and local authorities. It makes reporting easier and helps communities respond to problems faster and more effectively.

- **UN SDG Target**

The Community Issue Report System supports the goals of the United Nations (UN) SDG 11 “Sustainable Cities and Communities” by helping improve how local problems in the community are reported and managed by the organization like barangay.

Through the system residents can quickly report issues like loitering, flooding, and road damage, while barangay staff can organize and respond to these reports using priority levels and status tracking. This directly supports 11.3 by encouraging community participation and 11.6 by helping reduce environmental problems.

By making reporting faster, more organized and easier to track, the project helps create a cleaner, safer, and more responsive community.

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- **Connection of the System to the UN SDG**

The Community Issue Reporting System is directly connected to the goals of the United Nations SDG 11: Sustainable Cities and Communities because it helps solve everyday problems in the community in a more organized way.

The system allows residents to report issues like garbage, flooding, and damaged infrastructure, making it easier for local authorities to take action. Features like priority levels, status tracking, and report history help ensure that problems are handled properly and not ignored.

By improving communication between residents and barangay staff, the system supports a cleaner environment, safer public spaces, and better community management. This shows how technology can be used to achieve the goals of SDG 11 in a practical and effective way.

II. REQUIREMENT ANALYSIS

List of Functional and Non-Functional Requirements

This section presents the functional and non-functional requirements of the Community Issue Reporting System. These requirements define the system's

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expected behavior, features, and quality attributes necessary to ensure effective community issue reporting and management aligned with Sustainable Development Goal (SDG) 11.

A. Functional Requirements

Functional requirements specify the essential system functions that allow users and administrators to interact with the Community Issue Reporting System.

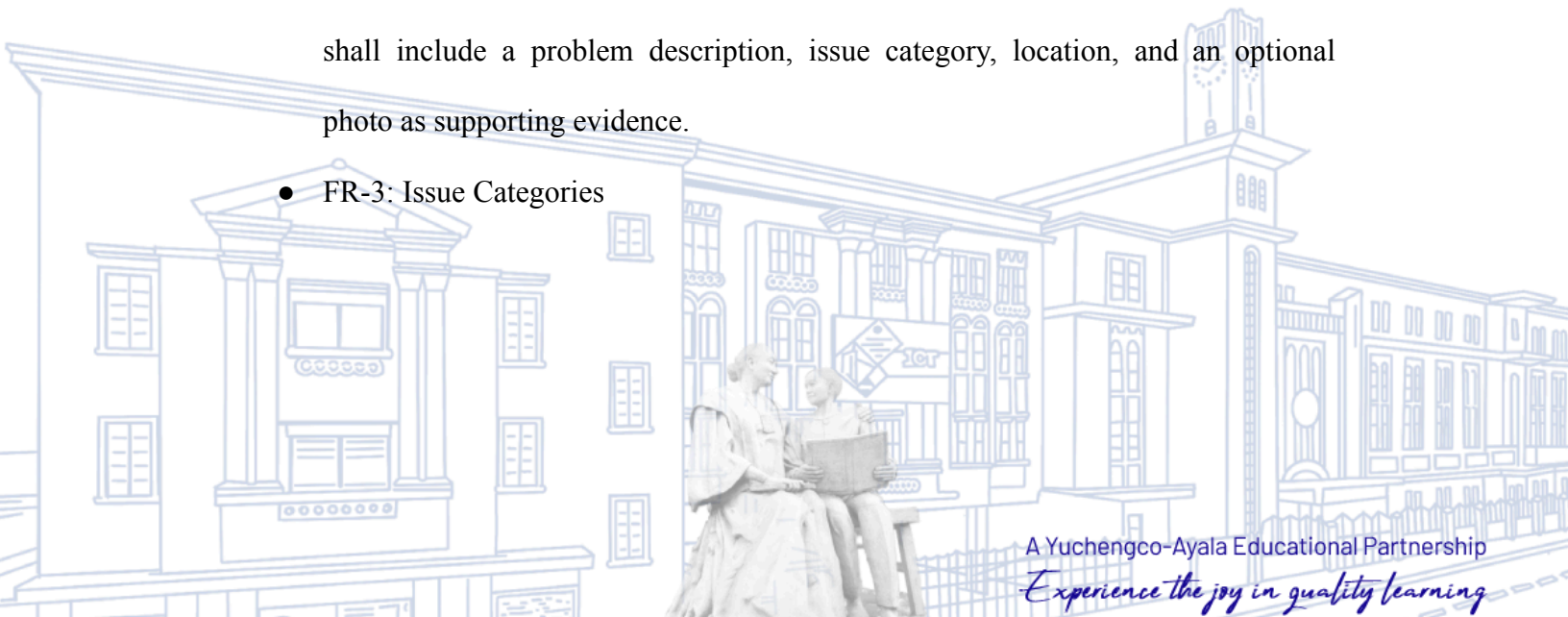
- FR-1: User Registration and Authentication

The system shall allow residents to register by providing valid user information. Registered users shall be able to log in and log out securely. Only authenticated users shall be permitted to submit and manage issue reports.

- FR-2: Issue Report Submission

The system shall enable users to submit community issue reports. Each report shall include a problem description, issue category, location, and an optional photo as supporting evidence.

- FR-3: Issue Categories



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The system shall support predefined issue categories, including but not limited to garbage disposal, road damage, flooding, street light malfunction, and noise complaints.

- FR-4: User Report Management

The system shall allow users to view their submitted reports. Users shall be able to edit or delete reports provided the report status remains pending.

- FR-5: Administrative Report Management

The system shall allow authorized barangay staff to access all submitted reports. Administrators shall be able to assign priority levels (Low, Medium, High, or Urgent) and update report statuses (Pending, In Progress, or Resolved).

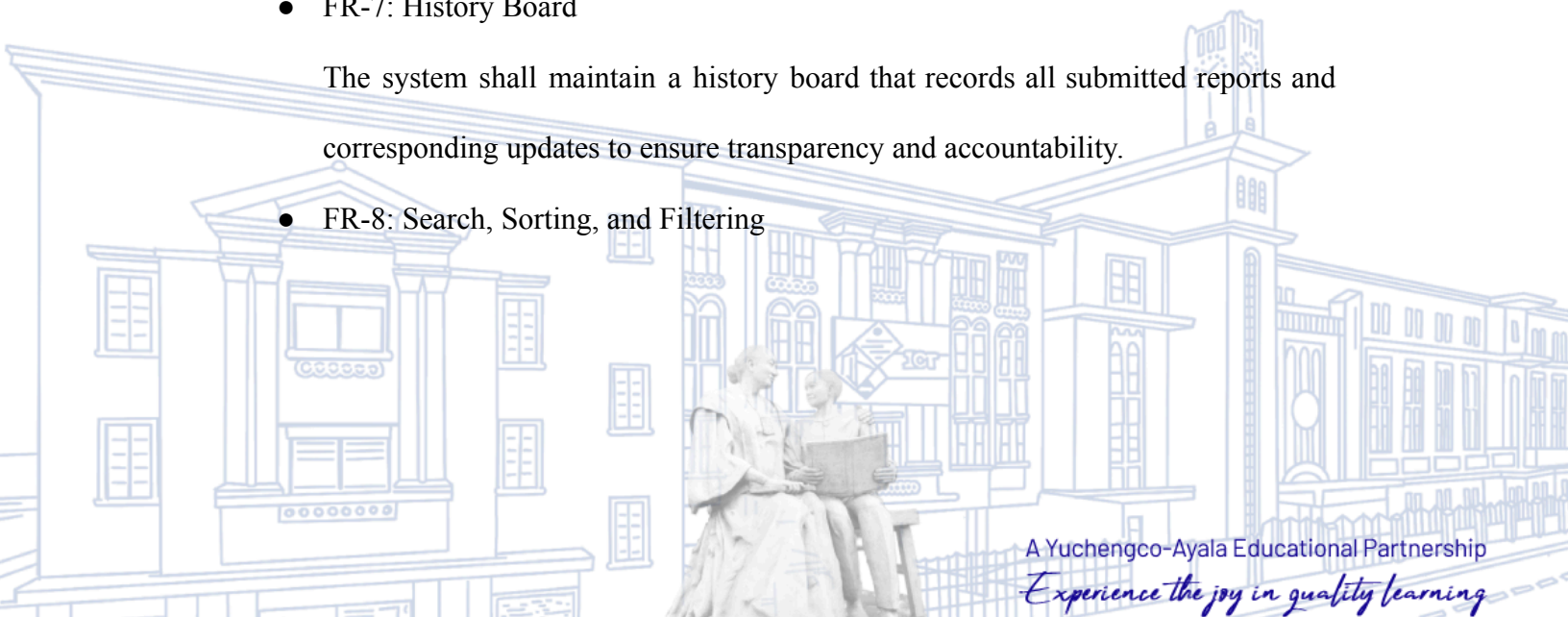
- FR-6: Assignment to Authorities

The system shall allow administrators to assign reported issues to appropriate barangay personnel or authorities for resolution.

- FR-7: History Board

The system shall maintain a history board that records all submitted reports and corresponding updates to ensure transparency and accountability.

- FR-8: Search, Sorting, and Filtering



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The system shall provide automatic sorting of reports based on category, priority level, status, and date. A search function shall be available to facilitate quick retrieval of reports.

- FR-9: Notification System

The system shall notify users when there are changes in the status or priority of their submitted reports.

- FR-10: Dashboard and Analytics

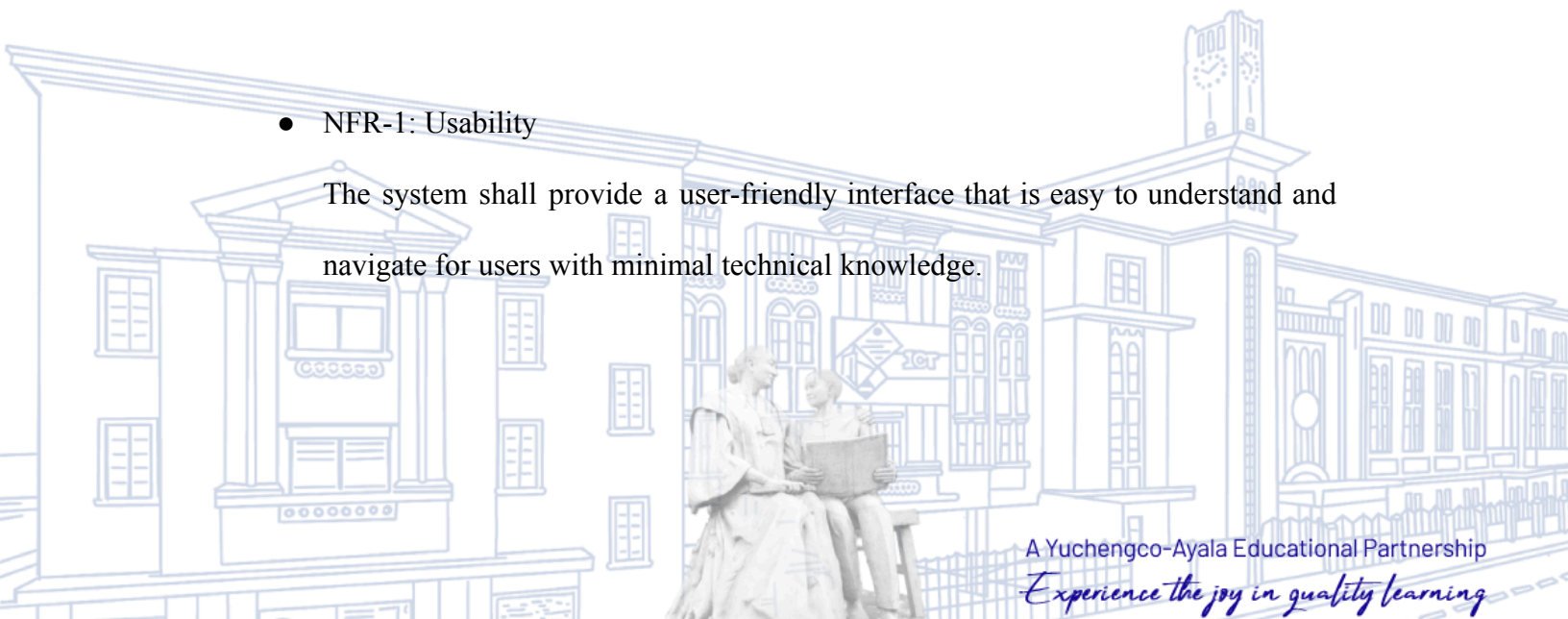
The system shall provide an administrative dashboard displaying the total number of reports, as well as counts of pending, in-progress, and resolved cases. Basic analytics showing the most frequently reported issues shall also be generated.

B. Non-Functional Requirements

Non-functional requirements describe the quality attributes and constraints under which the system must operate.

- NFR-1: Usability

The system shall provide a user-friendly interface that is easy to understand and navigate for users with minimal technical knowledge.



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- NFR-2: Performance

The system shall process user requests and load system pages within an acceptable response time under normal operating conditions.

- NFR-3: Reliability

The system shall ensure data accuracy and integrity. All submitted reports and user information shall be stored reliably without data loss.

- NFR-4: Security

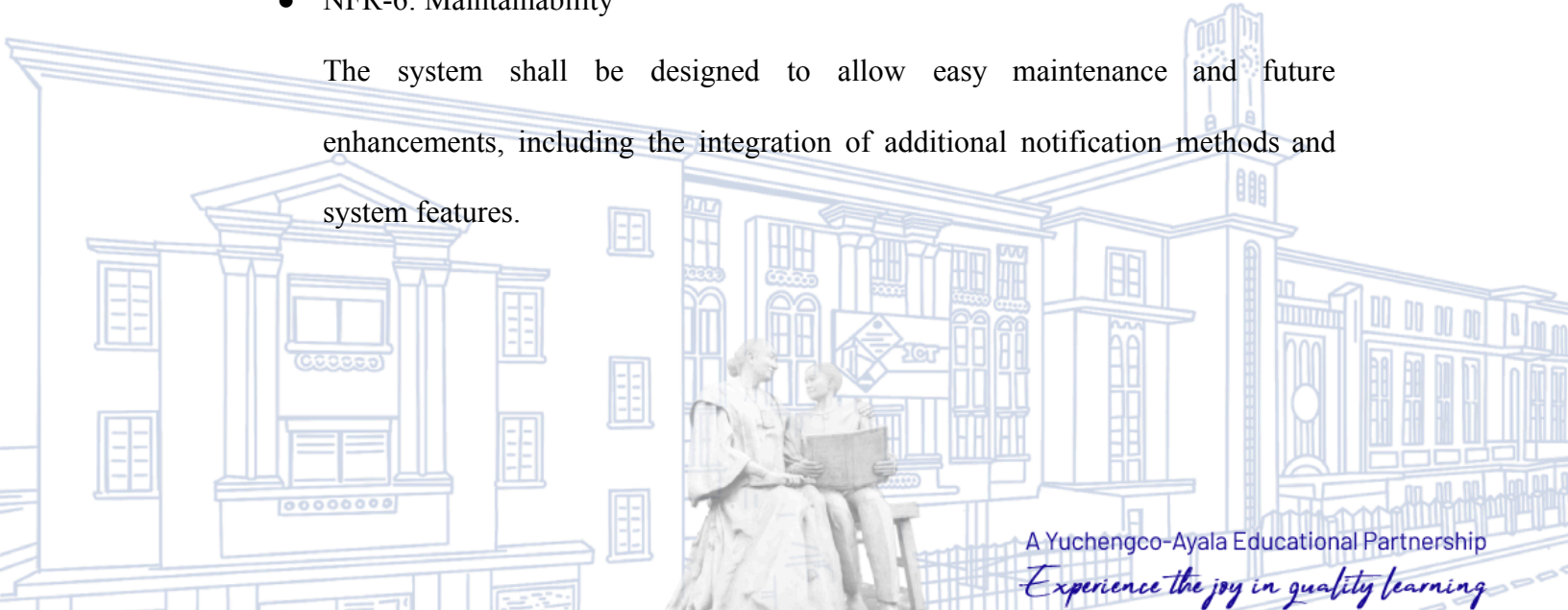
The system shall implement secure authentication mechanisms to protect user accounts. Access to administrative features shall be restricted to authorized personnel only.

- NFR-5: Scalability

The system shall be capable of accommodating an increasing number of users and reports as the community grows.

- NFR-6: Maintainability

The system shall be designed to allow easy maintenance and future enhancements, including the integration of additional notification methods and system features.



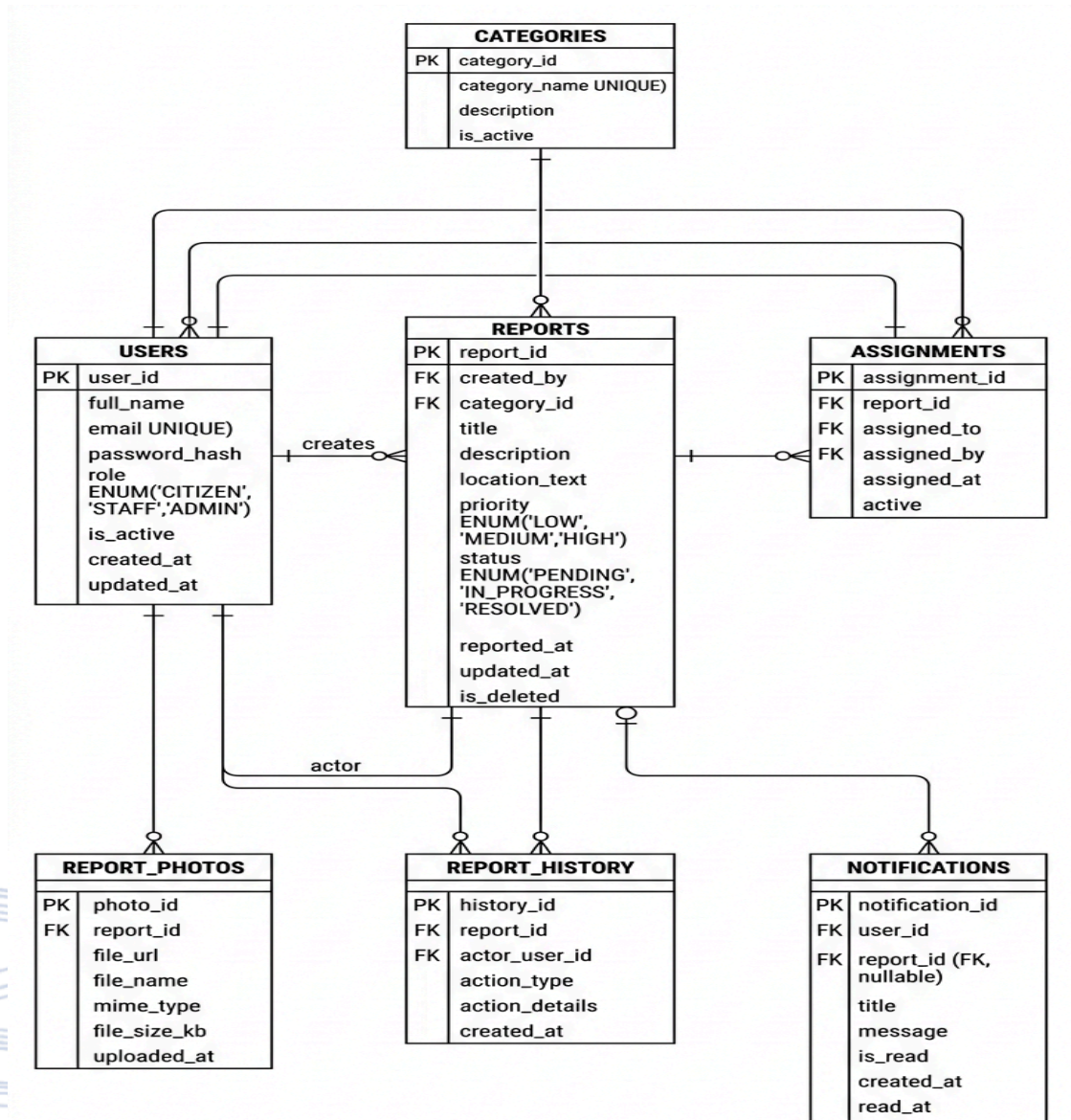
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III. DATABASE DESIGN

- Entity Relation Diagram (ERD)



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• Data Dictionary (Table Names, Fields, Types, and Constraints)

Table Name	Field Name	Data Type	Constraints	Description
USERS	user_id	INT	PK, AUTO_INCREMENT	Unique identifier for user
USERS	username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
USERS	email	VARCHAR(100)	UNIQUE, NOT NULL	User email address
USERS	password_hash	VARCHAR(255)	NOT NULL	Encrypted password
USERS	role	ENUM	NOT NULL, DEFAULT 'resident'	User role: resident or admin
USERS	phone	VARCHAR(20)	NULL	Contact phone number
USERS	created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp
USERS	updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Last update timestamp
REPORTS	report_id	INT	PK, AUTO_INCREMENT	Unique identifier for report
REPORTS	user_id	INT	FK → USERS(user_id)	Reference to user who submitted
REPORTS	category	ENUM	NOT NULL	Issue category: garbage, road_damage, flooding, street_lights, noise, riots
REPORTS	title	VARCHAR(200)	NOT NULL	Report title
REPORTS	description	TEXT	NULL	Detailed description
REPORTS	location_address	VARCHAR(255)	NULL	Location address text
REPORTS	latitude	DECIMAL(10,8)	NULL	GPS latitude
REPORTS	longitude	DECIMAL(11,8)	NULL	GPS longitude
REPORTS	photo_url	VARCHAR(500)	NULL	Photo file URL
REPORTS	priority_level	ENUM	DEFAULT 'low'	Priority: low/medium/high/urgent
REPORTS	status	ENUM	DEFAULT 'pending'	Status: pending/in_progress/resolved
REPORTS	assigned_to	INT	FK → USERS(user_id)	Admin assigned to resolve
REPORTS	created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Report creation timestamp
REPORTS	updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Last update timestamp
STATUS_HISTORY	history_id	INT	PK, AUTO_INCREMENT	Unique identifier for history entry
STATUS_HISTORY	report_id	INT	FK → REPORTS(report_id)	Reference to report
STATUS_HISTORY	old_status	VARCHAR(50)	NOT NULL	Previous status value
STATUS_HISTORY	new_status	VARCHAR(50)	NOT NULL	New status value
STATUS_HISTORY	changed_by	INT	FK → USERS(user_id)	User who made the change
STATUS_HISTORY	changed_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	When change occurred
STATUS_HISTORY	notes	TEXT	NULL	Additional notes
NOTIFICATIONS	notification_id	INT	PK, AUTO_INCREMENT	Unique identifier for notification
NOTIFICATIONS	user_id	INT	FK → USERS(user_id)	Recipient user
NOTIFICATIONS	report_id	INT	FK → REPORTS(report_id)	Related report
NOTIFICATIONS	type	ENUM	NOT NULL, DEFAULT 'push'	Notification type: email/sms/push
NOTIFICATIONS	message	TEXT	NOT NULL	Notification message content
NOTIFICATIONS	is_read	BOOLEAN	DEFAULT FALSE	Read status flag
NOTIFICATIONS	sent_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	When notification sent
HISTORY_BOARD	board_id	INT	PK, AUTO_INCREMENT	Unique identifier for board entry
HISTORY_BOARD	report_id	INT	FK → REPORTS(report_id)	Related report
HISTORY_BOARD	action_type	VARCHAR(50)	NOT NULL	Type of action performed
HISTORY_BOARD	action_by	INT	FK → USERS(user_id)	User who performed action
HISTORY_BOARD	action_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	When action occurred
HISTORY_BOARD	details	TEXT	NULL	Action details
ANALYTICS_LOG	log_id	INT	PK, AUTO_INCREMENT	Unique identifier for analytics entry
ANALYTICS_LOG	report_id	INT	FK → REPORTS(report_id)	Related report
ANALYTICS_LOG	category	VARCHAR(50)	NOT NULL	Issue category for analytics
ANALYTICS_LOG	priority	VARCHAR(20)	NOT NULL	Priority level for analytics
ANALYTICS_LOG	status	VARCHAR(20)	NOT NULL	Status for analytics
ANALYTICS_LOG	logged_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	When logged

Color Key: Primary Key (PK) Foreign Key (FK) USERS Table REPORTS Table NOTIFICATIONS ANALYTICS_LOG

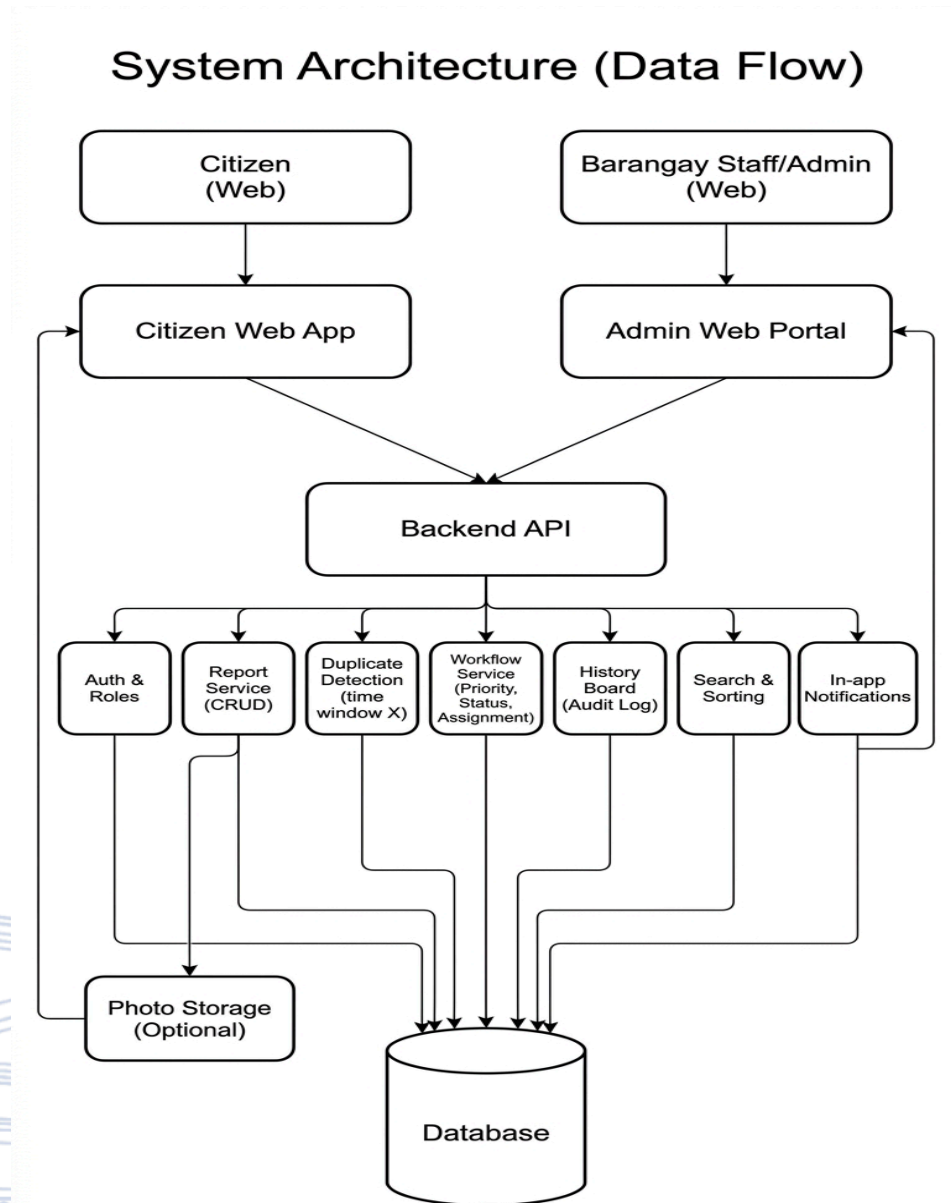
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IV. SYSTEM ARCHITECTURE

- Data Flow

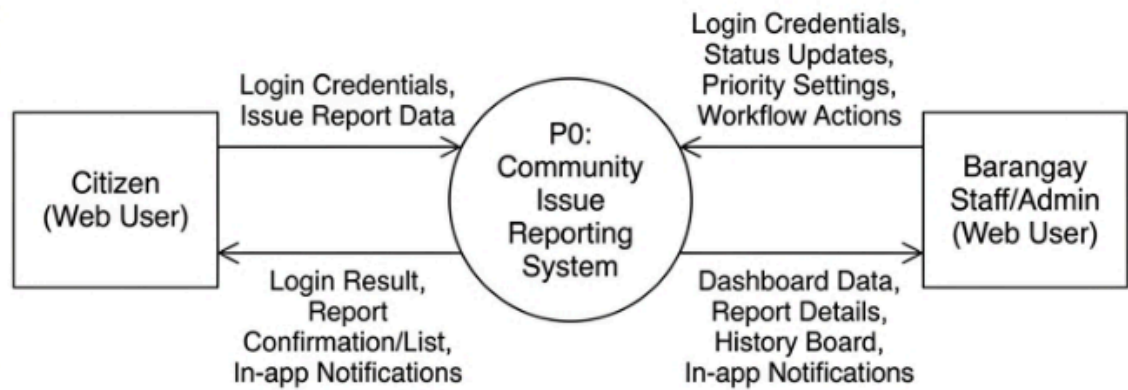




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- DFD Level 0 and 1

DFD Level 0



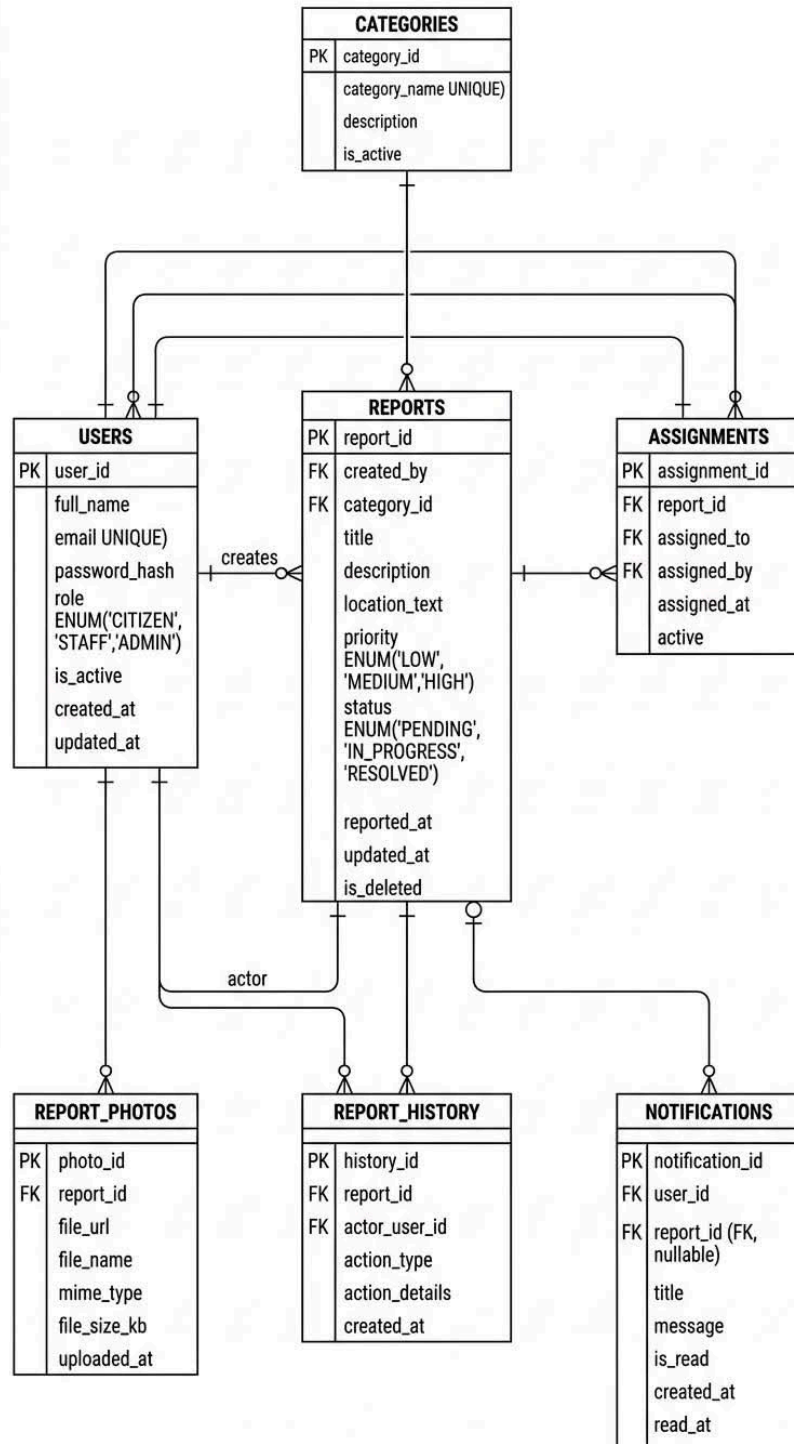
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V. INDIVIDUAL CONTRIBUTION

Wireframe & User Interface – Vince Alex M. Alcala & Junior C. Delos Santos

System Flow & Code – Renz Christian M. Malco & Rehan T. Alamada

Documentation – Sophia Bea U. Rivera & Lance Andrei A. Gulila

IV. AI-ASSISTED TOOLS USED IN THE STUDY

- ChatGPT (OpenAI) – Assisted in system design guidance, CRUD logic explanation, debugging support, and documentation clarity.
- Gemini (Google) – Used for information gathering and concept verification.
- Grammarly (Grammarly, Inc.) – Used for grammar checking and sentence improvement.

